A GUIDED EXERCISE IN READING EVIDENCE

Mapping the ageing stromal niche

An original teaching example for connecting scientific questions, figure panels and evidence.

THE QUESTION

Does an inflammatory stromal state accompany ageing, and what would it take to test its function?

Abstract

Tissue ageing can be described in terms of cell composition, within-cell state and functional interactions. These are related but distinct questions. In this fictional case, a single-cell atlas motivates an association analysis and a subsequent intervention. The aim is to practise matching the strength of each claim to the experiment that supports it.

A six-step reading route

Question	Design	Atlas	Association	Perturbation	Boundaries
01	02	03	04	05	06

What to look for

Experimental unit Patients, donors or animals provide biological replication.

Axes and units Identify what each number represents before interpreting a trend.

Evidence boundary Distinguish a descriptive pattern from a causal intervention.

A DEMONSTRATION, NOT A SCIENTIFIC FINDING

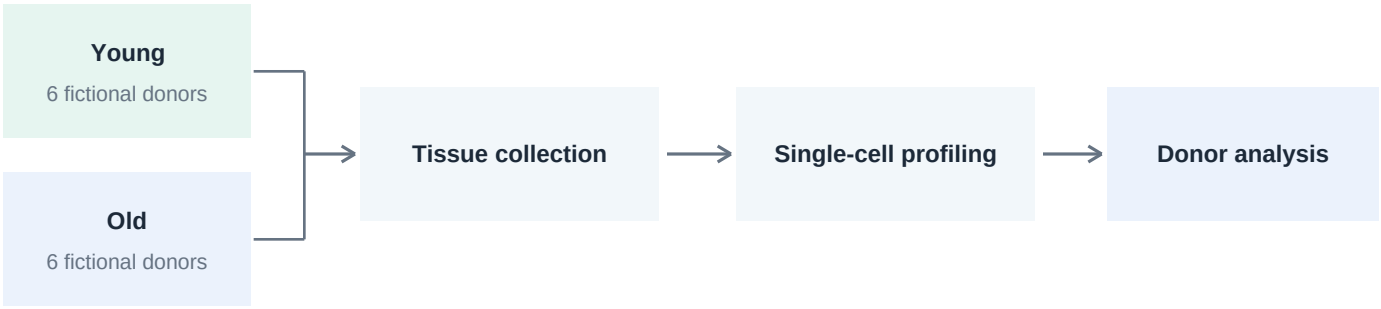
All samples, values, figure panels and conclusions in this document are invented. No study was performed. Do not cite this material as biological evidence.

01 / STUDY DESIGN

A sample is not a cell

Before following the results, locate the biological replicates and examine what is being compared.

A Two cohorts, one sample-level comparison



B Recovered cells by donor

HOW TO READ FIGURE 1

Each point in B is one fictional donor. Similar cell recovery does not establish similar viability, absence of batch effects, or equal cell-type abundance.

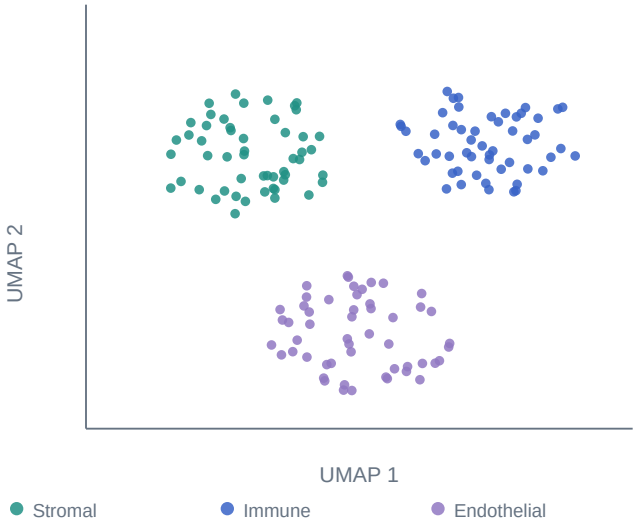
Figure 1 | Study design and sample-level quality control. All values are synthetic.

02 / ATLAS OBSERVATIONS

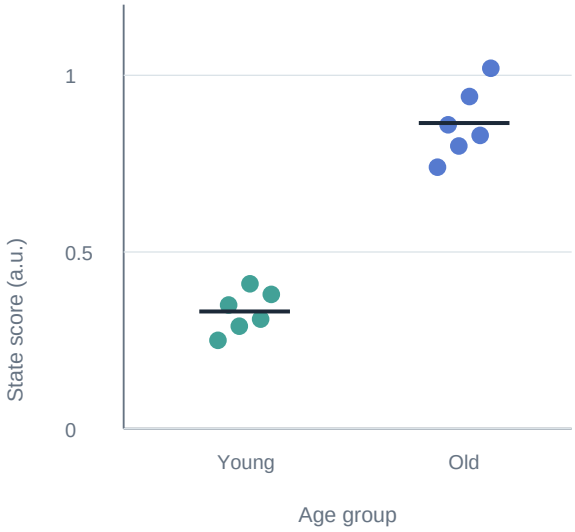
Separate composition from state

A cell map, a donor-level score and a marker heatmap answer three different questions.

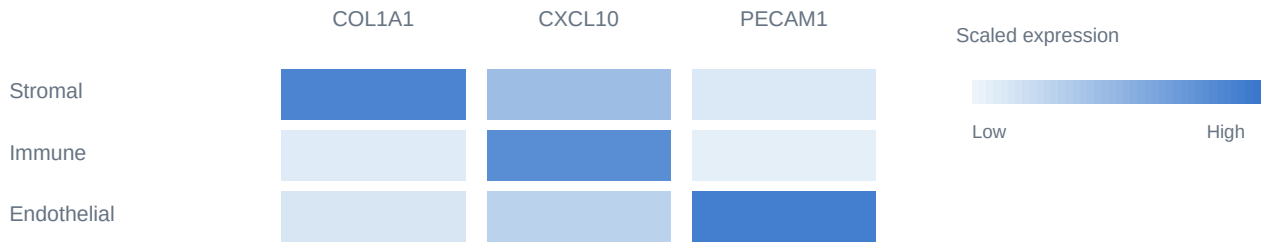
A **Illustrative cell atlas**



B **Inflammatory state**



C **Marker expression (illustrative)**



HOW TO READ FIGURE 2

A: coordinates are dimensionless. B: six points are six donors per group. C: color is relative expression, not absolute abundance. No p-values or real effect sizes were computed.

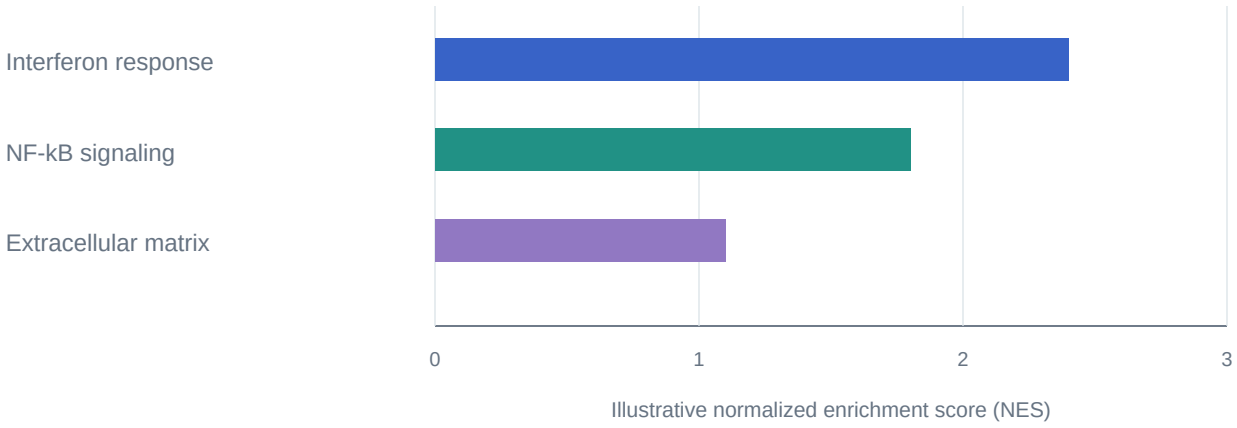
Figure 2 | An illustrative cell atlas. Every coordinate and expression value is invented.

03 / MECHANISTIC CLUES

Association suggests a test

Pathway-level patterns and cross-cell associations help formulate hypotheses for intervention.

A Candidate pathway signals



B Scores within the same donor

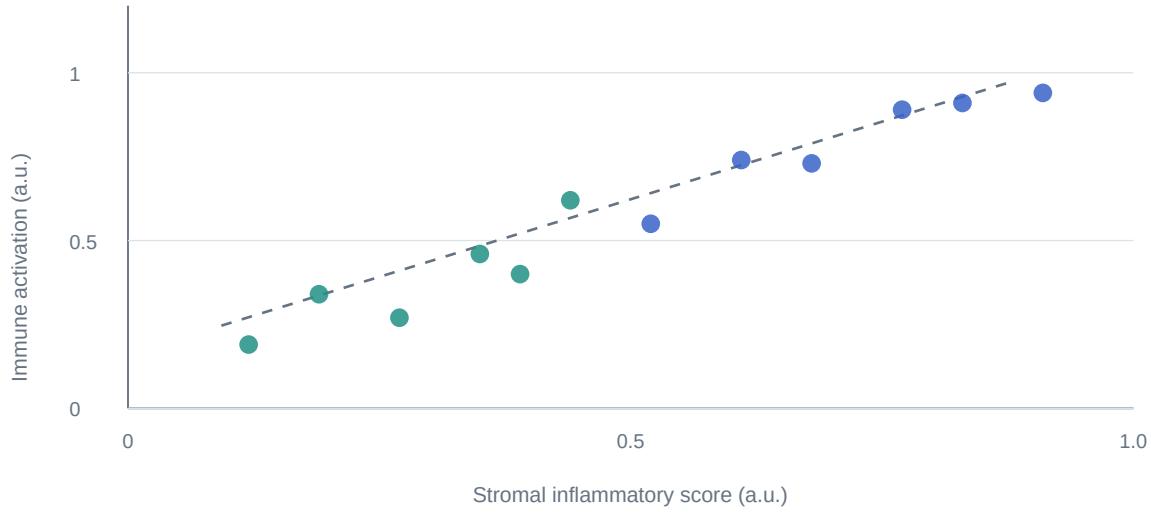


Figure 3 | Synthetic values. No enrichment, correlation or regression was performed. The dashed line is a visual guide, not an estimated regression.

04 / FUNCTIONAL EXPERIMENT

Match the intervention to the claim

A perturbation and a rescue can clarify function only when the relevant controls are present.

A Conditioned-medium transfer

Stromal perturbation
Modify candidate process

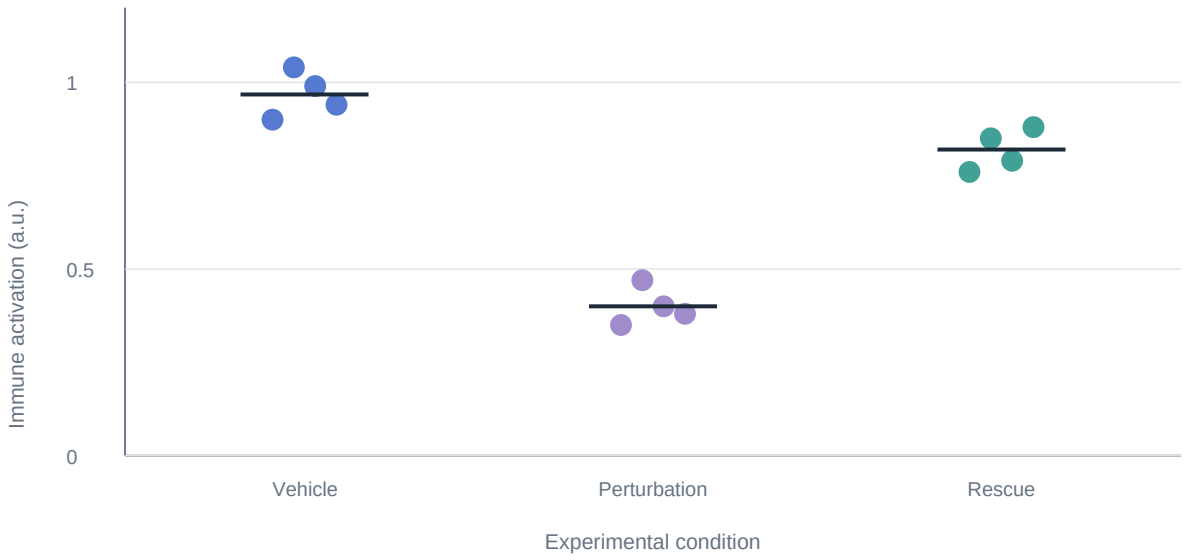
→

Medium transfer
Test transferable effects

→

Immune readout
Measure functional response

B Illustrative intervention and rescue



CONTROLS MAKE THE INTERPRETATION

Check cell viability, drug carry-over, target engagement, dose and specificity. Four dots per condition represent four invented biological replicates, not technical wells.

Figure 4 | A fictional intervention, shown only to teach experimental logic. No experiment was performed and no statistical inference is claimed.

05 / EVIDENCE BOUNDARIES

What can the data support?

A useful reading note records both the finding and the experiment that would change your confidence.

QUESTION	EVIDENCE TO LOOK FOR	LIMIT
Composition	Donor-level proportions	Cell-map density alone is insufficient.
Within-cell state	Donor-level aggregation	More cells do not replace more donors.
Mechanism	Target engagement + rescue	Association does not show direction.
Translation	Independent human validation	A model does not establish clinical benefit.

A five-question reading checklist

1. What is the experimental unit?
2. What do the axes and normalization mean?
3. Which exact panel supports this sentence?
4. Is the result descriptive, associative or interventional?
5. Which controls or methods are missing?

END OF EDUCATIONAL DEMONSTRATION

Upload a real paper and verify every interpretation against the original PDF.